Embedded-AI PCB for Predictive Bearing-Fault Diagnosis in Industrial Rotating Machinery via Vibration Analysis

1. Project title and team

Team: ITESCA PCB Lab · **Institution:** ITESCA (Instituto Tecnológico Superior de Cajeme) · **Advisor:** Dr. Marcos Alan Cota Leal

Members: Juan Alfonso Moroyoqui Delgadillo (Computer Systems Engineering) · Carlos Eduardo Tamayo Urrea (Computer Systems Engineering) · Alex Hiram Meza Esquer (Computer Systems Engineering)

Contact: juanmorodel@gmail.com · carlostamayodev@gmail.com · alexhirameza@gmail.com

2. Purpose and industrial context

Bearing faults account for roughly 40% of induction-motor failures in classic reliability surveys [1] and are a leading source of unplanned stops in rotating machinery; our board detects them on the machine itself, in real time, at a fraction of the cost of a SCADA-based monitoring system.

Sector and need. The project targets industrial automation and predictive maintenance (Industry 4.0): motors, pumps, fans and compressors in plants and small and medium enterprises (SMEs) that cannot afford permanent condition-monitoring systems. A dedicated printed circuit board assembly (PCBA) puts the sensor, classifier and wireless link on one 5 × 5 cm board on the motor housing; only the diagnosis leaves the device.

Expected behavior. The board acquires a vibration window, computes 22 features, runs a small neural network and reports the bearing state with a confidence level. It must work unattended and keep working in a harsh environment:

- **Power:** 9–12 V DC input from an industrial supply, stepped down to 3.3 V by an LM2596 switching regulator.
- **Temperature:** every main component is rated for –40 °C up to at least +85 °C, enough for a motor housing.
- **Real-time constraint:** a 1024-sample window at 13.3 kHz lasts about 77 ms; the full feature extraction and inference must finish well inside that time on a 100 MHz Cortex-M4F core.

3. PCB architecture and design cycle (Proteus)

3.1 Hardware architecture. Data flows from the power input, through regulation, to the sensor, the microcontroller and the wireless link:

- **Power:** a 9–12 V DC input passes a reverse-polarity diode into an LM2596S-3.3 step-down converter (3 A, fixed 150 kHz) with a Schottky catch diode, inductor and capacitors, producing the 3.3 V rail.
- **Input conditioning:** the IIS3DWB is a digital, low-noise vibration sensor, so there is no analog front end to pick up noise: it samples three axes at 26.667 kHz (bandwidth about 6 kHz) and sends them over a 4-wire SPI bus (PA4–PA7); its INT1 data-ready output goes to PA1.
- **Microcontroller:** an STM32F411CE (Cortex-M4F at 100 MHz, floating-point unit and DSP instructions, 512 KB flash, 128 KB RAM) runs the FFT and the neural network; it was chosen because its floating-point unit and DSP instructions make both fast enough at low cost.
- **Communication:** an ESP32-C3-MINI-1 module (Wi-Fi/Bluetooth, pre-certified) receives the diagnosis over USART1; headers expose SWD debug, a UART and a test pin. There are no actuators: the board's output is the diagnosis itself.

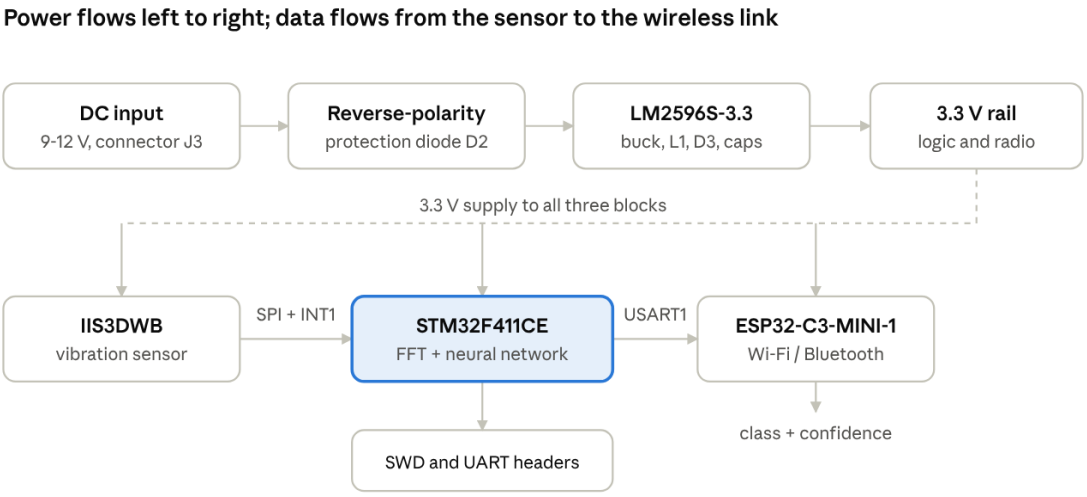


Figure 1 · block diagram: power chain, sensing, processing and wireless link

3.2 Design and verification in Proteus. We captured the schematic, built a custom LGA-14 footprint and symbol for the IIS3DWB from its datasheet and the ST soldering guideline TN0018, placed all parts on the top layer, and filled the bottom layer with a ground-plane copper pour created with the Power Plane Generator. The Proteus checks all pass: 0 unrouted connections, 0 connectivity errors, 0 design-rule (DRC) errors (clearances: pad-pad 0.127 mm, track-track 0.152 mm, copper to edge 0.38 mm) and a successful Pre-Production Check. The classifier chain (22 features plus the network) was verified in Python with the exported model; the 12 V to 3.3 V power stage was also simulated in Proteus (Figure 2, Annex G).

The LM2596S-3.3 has no Proteus simulation model, so we simulated an equivalent buck converter with the same inductor (33 µH), input capacitor (680 µF) and output capacitor (470 µF), a generic Schottky diode, a generic P-channel MOSFET and a 33 Ω load (about 100 mA). An open-loop 150 kHz PWM source (22 % duty cycle) replaces the regulator's internal control loop. The output rises smoothly without overshoot and settles at about 3.3 V (3.28 V in the animated run, within 0.5 % of the target). At this light load the inductor runs in discontinuous conduction mode, so the duty cycle was tuned by hand; the real regulator corrects this through its feedback loop. The firmware was then run on the STM32F411CE of our board in Proteus VSM (Virtual System Modelling): the program is loaded into the simulated microcontroller and each diagnosis is printed on a Virtual Terminal connected to USART1 (PA9, 115200 baud), 16 of 16 test windows correct (Section 4.3). Proteus does not model the Cortex-M4 floating-point unit, so that run used a software-float build.

3.3 Standards and noise mitigation.

- **Layout critical points:** the accelerometer sits about 1.5 cm from the microcontroller so the SPI lines stay short; the switching stage (converter, inductor, catch diode) is in the opposite lower corner, about 2.5 cm from the sensor, with a compact switching loop; the radio antenna is at the board edge with a copper keepout on all layers.

- **Power handling:** power tracks are wider (0.5–0.76 mm) than signal tracks (approximately 0.25 mm). Per IPC-2221, a 0.5 mm track in 1 oz copper carries about 1.5 A at a 10 °C rise, far above the roughly 0.5 A the board draws.
- **Signal integrity and EMI:** a ground-plane copper pour covering most of the bottom layer (a few signal tracks cross it), a decoupling capacitor on every STM32 VDD and VDDA pin and on VCAP (VBAT is tied directly to 3.3 V), plus two capacitors next to the sensor, short SPI traces at a few MHz (no impedance control needed), and the unused interrupt pin INT2 left open by design.
- **Standards:** IPC-2221 (generic PCB design: spacing, track width) and IPC-7351 (land patterns) guided the design rules and footprints.

4. Artificial intelligence / machine learning integration

4.1 Objective. The AI optimizes the operation of the PCBA (fault diagnosis), not the physical design process of the board.

4.2 Dataset. We use the Case Western Reserve University (CWRU) Bearing Data Center, drive-end accelerometer at 12 kHz, 0.007-inch faults, four motor loads (0–3 HP). Each signal is resampled to 13.33 kHz (half of the IIS3DWB output rate, 26.667 kHz) and cut into 1024-sample windows with 50% overlap, giving 6,837 windows. Training uses loads 0–2 HP; testing uses 3 HP, a condition the model never saw.

- **Input (22 features per window):** root mean square (RMS), peak, crest factor, kurtosis, skewness, standard deviation, and the base-10 logarithm of the energy in 16 frequency bands of a Hann-windowed fast Fourier transform (FFT).

4.3 Model. A multilayer perceptron (MLP) with layers 22 → 32 → 16 → 4 (ReLU activations, softmax output), trained with the Adam optimizer for 40 epochs and class weights to balance the healthy class. It has 1,332 parameters (about 5 KB in float32); the exported TensorFlow Lite file is 8 KB. The same 22 features and the network were implemented in C and run on a simulated STM32F411CE (Proteus VSM): 16 of 16 test windows were diagnosed correctly, and 9,215 windows gave predictions identical to the Python model. Timing on the physical board is pending.

4.4 Results. On the 3 HP test load the classifier scores 100% on 1,843 clean windows (precision, recall and F1 of 1.00 in every class), and leave-one-load-out validation gives 100% for each held-out load. It also holds up under noise and gain changes (table in Section 5).

SNR is the signal-to-noise ratio. Results are laboratory data; with the real IIS3DWB the model must be validated again on a test rig (Section 7).

5. AI-based optimization of the PCBA: before vs. after

5.1 System level. Without the AI, the board would only be a data logger: three axes at 26.667 kHz and 16 bits produce about 160 KB/s that an external computer or SCADA system must analyze. With the embedded model, the microcontroller sends one class and a confidence value per 77 ms window, so a low-power Wi-Fi/Bluetooth link is enough and the decision is made locally, with no cloud dependency.

5.2 Model level. Our first model (a baseline trained on CWRU at 12 kHz) scored 100% on clean laboratory data, but stress tests against the real board exposed four problems, each now fixed: a

sampling-rate mismatch (we now resample to the IIS3DWB's 13.33 kHz), noise fragility (we switched to log-scale band energies and added noise and gain augmentation), a data-loading bug and class imbalance (we added class weights).

Test on the 3 HP load, simulated for the real sensor	Before	After
Clean data at 12 kHz	100%	100%
Signal at sensor rate 26.667 kHz, no adjustment	71%	not used (13.33 kHz)
Noise SNR 20 dB	14%	100%
Noise SNR 5 dB	14%	100%
Noise SNR 0 dB	14%	about 99%
Gain ×0.5	44%	100%
Gain ×2	88%	99.9 %

The "before" column was measured by reproducing the baseline with an equivalent network; the "after" column is the final TensorFlow model.

6. Economic analysis and commercial viability

We estimate the electronics at roughly US$17–41 per board in prototype quantities, against about US$275 per sensor node for a commercial wireless vibration kit.

6.1 Component cost (per board, prototype quantities). Prices are estimates in US dollars: the microcontroller price was checked at the distributor; the rest are ranges to be confirmed with quotes.

Item	Part	Estimated cost (USD)
Microcontroller	STM32F411CE (from $3.73 at LCSC)	3.7 – 8
Vibration sensor	IIS3DWB digital accelerometer	5 – 10
Wireless module	ESP32-C3-MINI-1 (Wi-Fi/Bluetooth)	5 – 15
Power stage and passives	LM2596S-3.3, inductor, diodes, capacitors, resistors	2 – 5
Connectors	Power, debug (SWD), UART	1 – 3
Total components		**≈ 17 – 41**

A mechanical enclosure is not included in the prototype; a sealed plastic case would add an estimated US$5–10.

6.2 Fabrication and assembly estimate. A 2-layer 5 × 5 cm board needs no special process. For a prototype batch we estimate US$20–50 for the bare boards. JLCPCB's published assembly fees show that soldering is cheap: about US$0.0016 per solder joint, under US$0.50 for a board with 250 joints. Our working hypothesis for 1,000+ units is bare boards in the order of US$1–3 each plus assembly, which would bring a finished board to roughly US$20–35 including components. This hypothesis must be confirmed with a volume quote.

6.3 Market comparison. Commercial condition-monitoring systems are priced for large plants. The Fluke 3561 FC wireless vibration sensor starter kit (four sensors plus one gateway) is listed from about US$1,100, roughly US$275 per sensor, with an optional software subscription. Our board performs the diagnosis locally on the device.

6.4 Commercial viability.

- **Pros:** an order of magnitude cheaper than commercial kits; no cloud or subscription; compact 5 × 5 cm board; reuses one proven design for every motor.
- **Cons:** the classifier is trained on laboratory data and needs validation on real machines; only four fault classes, with no severity estimate; no industrial certification yet (EMC, ingress protection).
- **Scalability:** a standard 2-layer process and widely available parts allow fabrication in volume. A target selling price of US$60–120 per node would leave a healthy margin over the estimated US$25–45 cost including enclosure.

7. Conclusions

We delivered a routed, rule-clean 2-layer PCB and a validated 8 KB classifier that together diagnose bearing faults on the machine, with the main remaining risk being validation on a real motor.

- **Main contribution and innovation:** a compact 5 × 5 cm board that unites a digital vibration sensor, a DSP-capable microcontroller and a wireless module, plus an embedded model whose features and sampling rate were deliberately matched to the board's own sensor (13.33 kHz) and hardened against noise and gain variation.
- **Limitations:** the model was trained and tested on laboratory data (CWRU) with a single fault size and a different accelerometer; no physical prototype has been built (not required by the competition); the Proteus firmware run uses a corrected IIS3DWB symbol in a simulation copy and a software-float build, and timing on real hardware is not measured; the power-stage simulation uses generic components and open-loop control instead of the LM2596S model, and closed-loop behavior under load steps is not simulated; four fault classes only, without severity.
- **Industrial potential:** at an estimated US$25–45 per node, against roughly US$275 for a commercial kit, the design could bring predictive maintenance to SMEs.
- **Next steps toward production:** (1) port the IIS3DWB SPI driver and the ESP32-C3 firmware, and measure inference time on the board; (2) fabricate a prototype batch; (3) validate on a motor test rig with real faults and retrain; (4) run electromagnetic compatibility tests and design the enclosure.

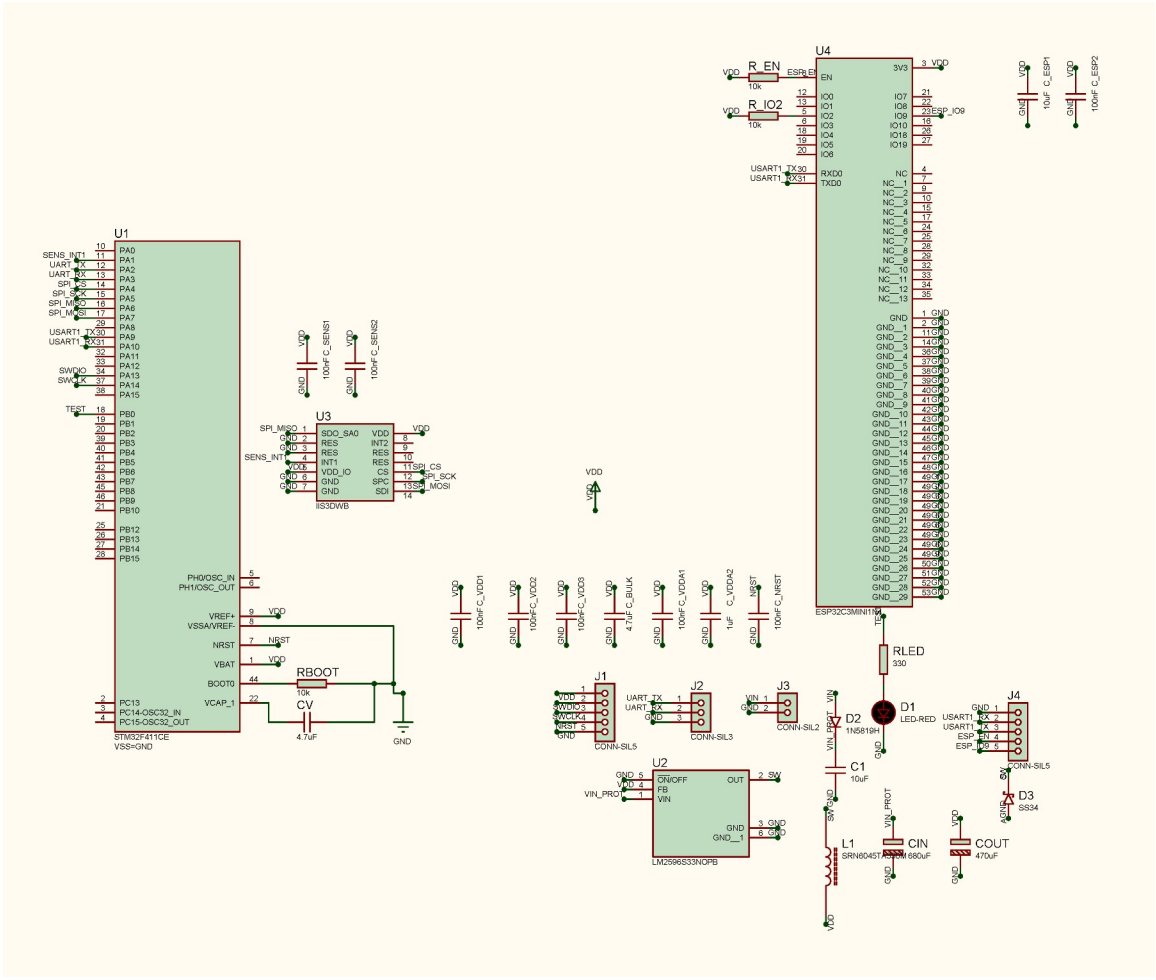
8. References

1. P. F. Albrecht et al., "Assessment of the reliability of motors in utility applications – Updated," *IEEE Trans. Energy Convers.*, vol. EC-1, no. 1, pp. 39–46, Mar. 1986.
2. W. A. Smith and R. B. Randall, "Rolling element bearing diagnostics using the Case Western Reserve University data: A benchmark study," *Mech. Syst. Signal Process.*, vol. 64–65, pp. 100–131, 2015.
3. Case Western Reserve University, Bearing Data Center. engineering.case.edu/bearingdatacenter
4. STMicroelectronics, *IIS3DWB: Ultrawide bandwidth, low-noise, 3-axis digital vibration sensor*, datasheet.
5. STMicroelectronics, *STM32F411xC/xE datasheet*.
6. Texas Instruments, *LM2596 SIMPLE SWITCHER 150-kHz 3-A step-down voltage regulator*, datasheet.
7. Espressif Systems, *ESP32-C3-MINI-1 datasheet*.
8. STMicroelectronics, *TN0018: Handling, mounting and soldering guidelines for MEMS devices*.
9. IPC, *IPC-2221: Generic Standard on Printed Board Design*; *IPC-7351: Generic Requirements for Surface Mount Design and Land Pattern Standard*.
10. JLCPCB, PCB assembly price (accessed Nov. 7, 2026).
11. LCSC, STM32F411CEU6 listing (accessed Nov. 7, 2026).
12. Instrumart, Fluke 3561 FC vibration sensor starter kit (accessed Nov. 7, 2026).

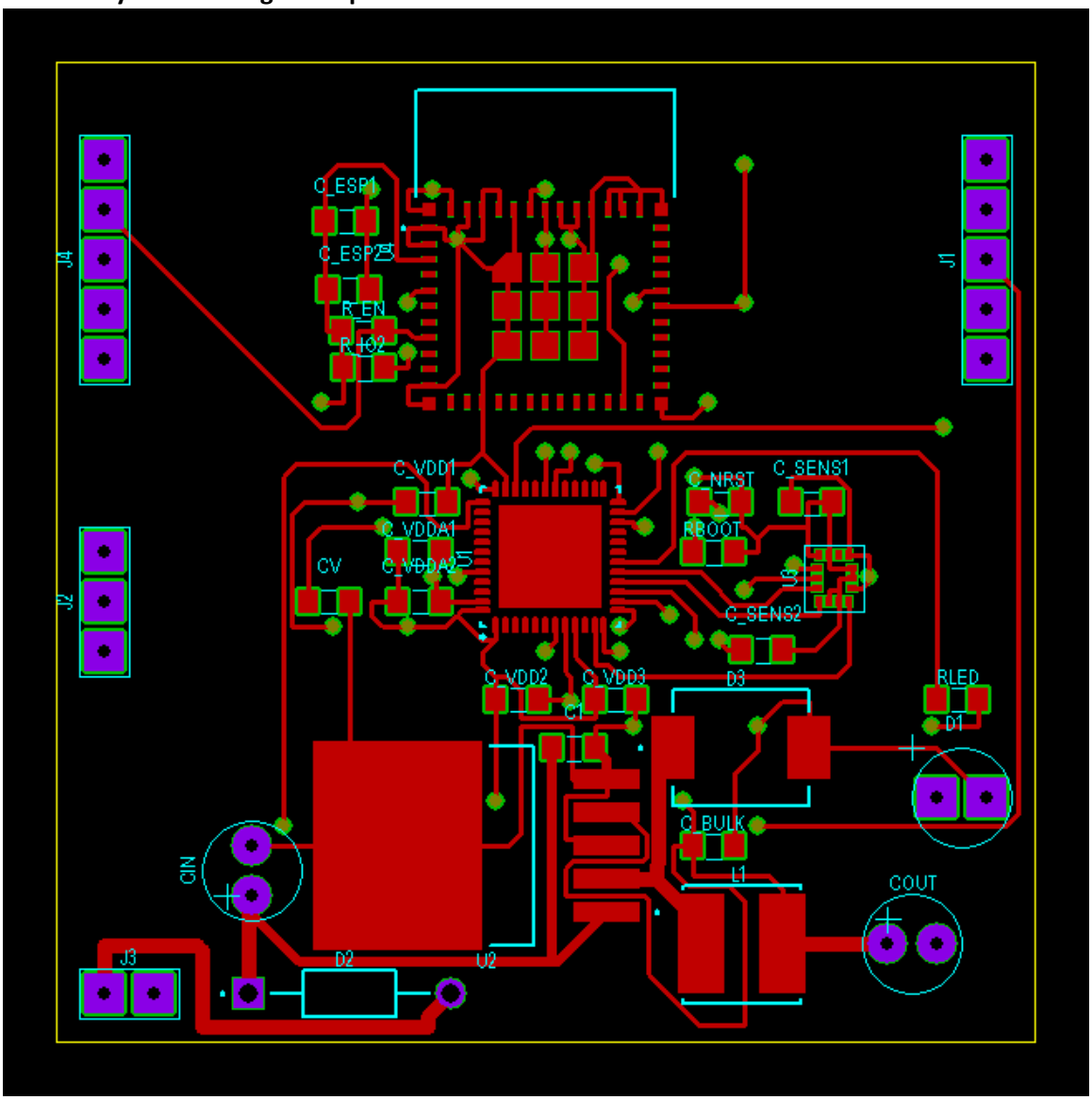
9. Annexes

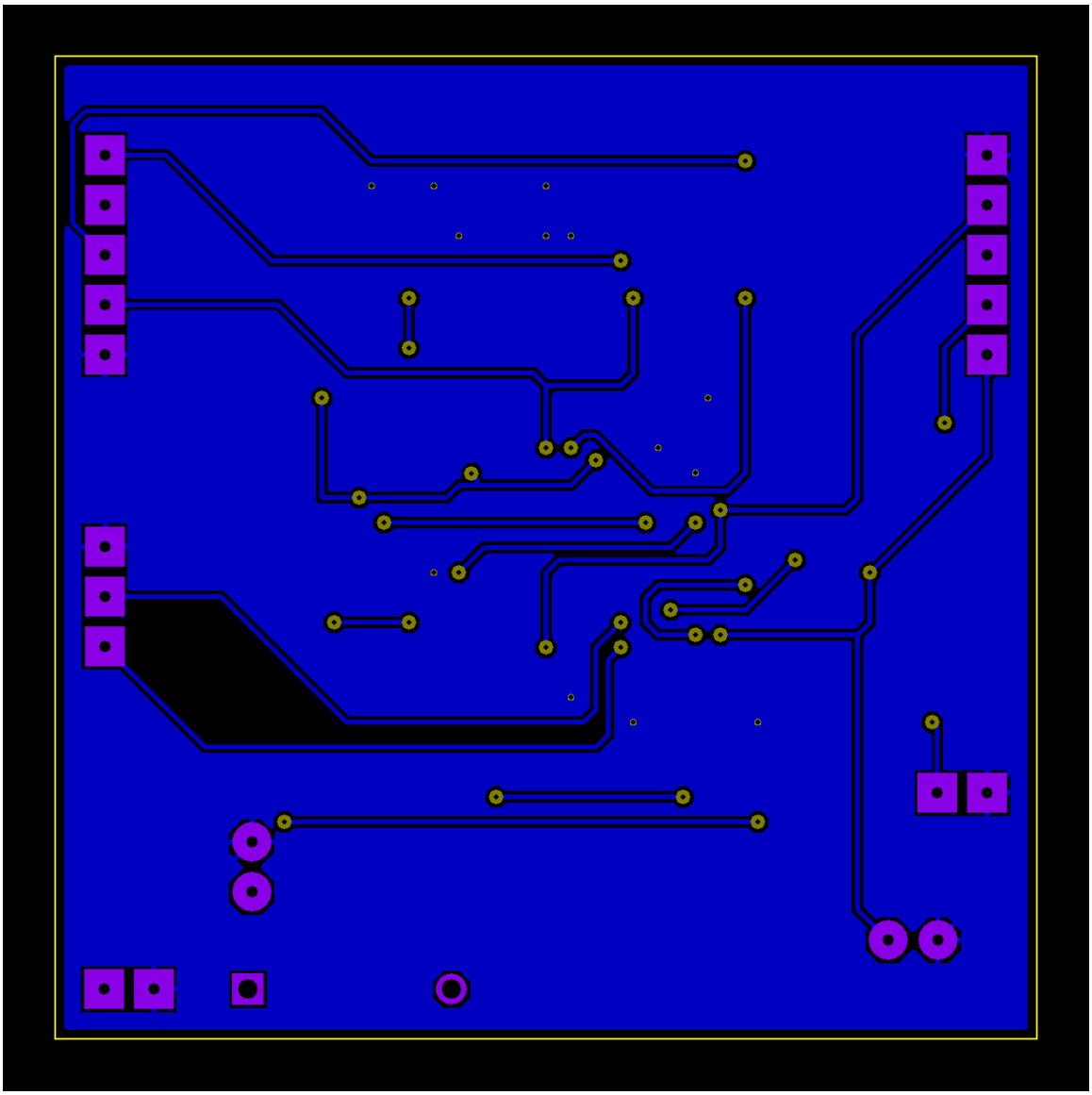
- **A. Source code and design files:** github.com/Beelzemon97/AI-SED-PCB (public). Proteus project in `hardware/proteus`, IIS3DWB library in `hardware/library`, training script `model/modeloCWRU_corregido.py`, exported model `model/modelo_rodamientos_corregido.tflite`, Gerber files in hardware/gerber, the C inference pipeline in firmware/src, the demo firmware in firmware/app, the bill of materials BOM_AI-SED-PCB.csv in hardware/bom, and the Proteus simulation copy in hardware/proteus and C weights `model/modelo_pesos.h`.

- **B. Full Proteus schematic:**



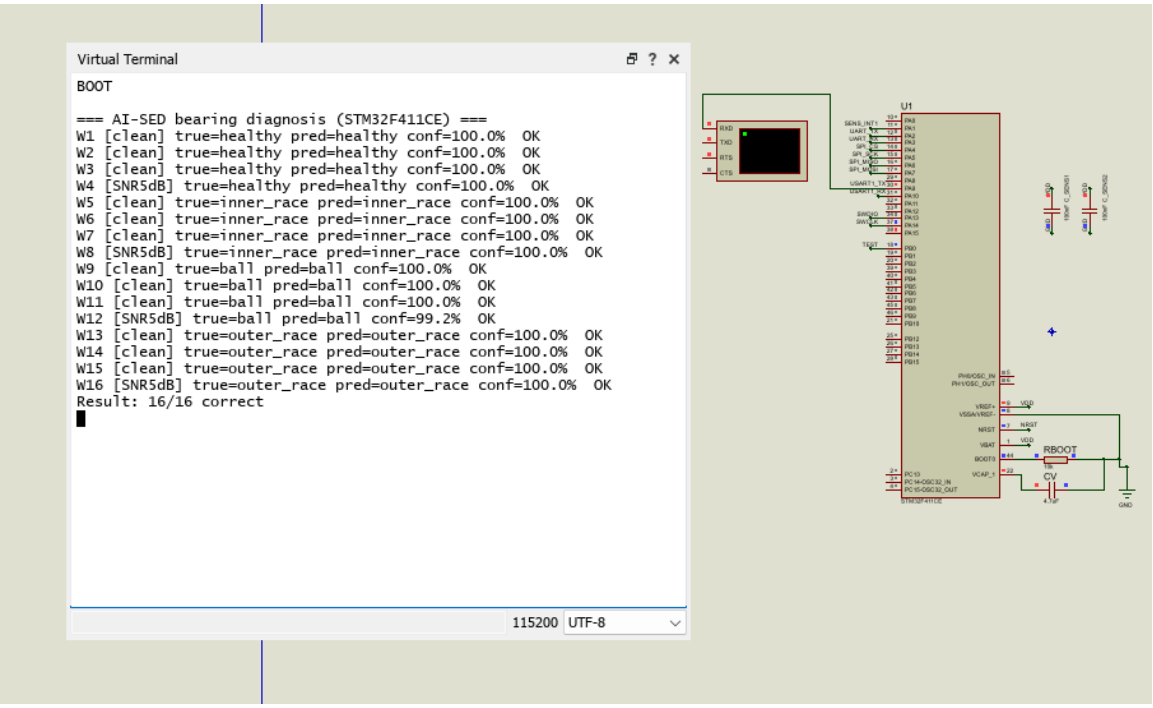
- **C. PCB layout views (ARES, 5 × 5 cm): top layer with copper, silkscreen and board edge, then bottom layer with the ground plane.**



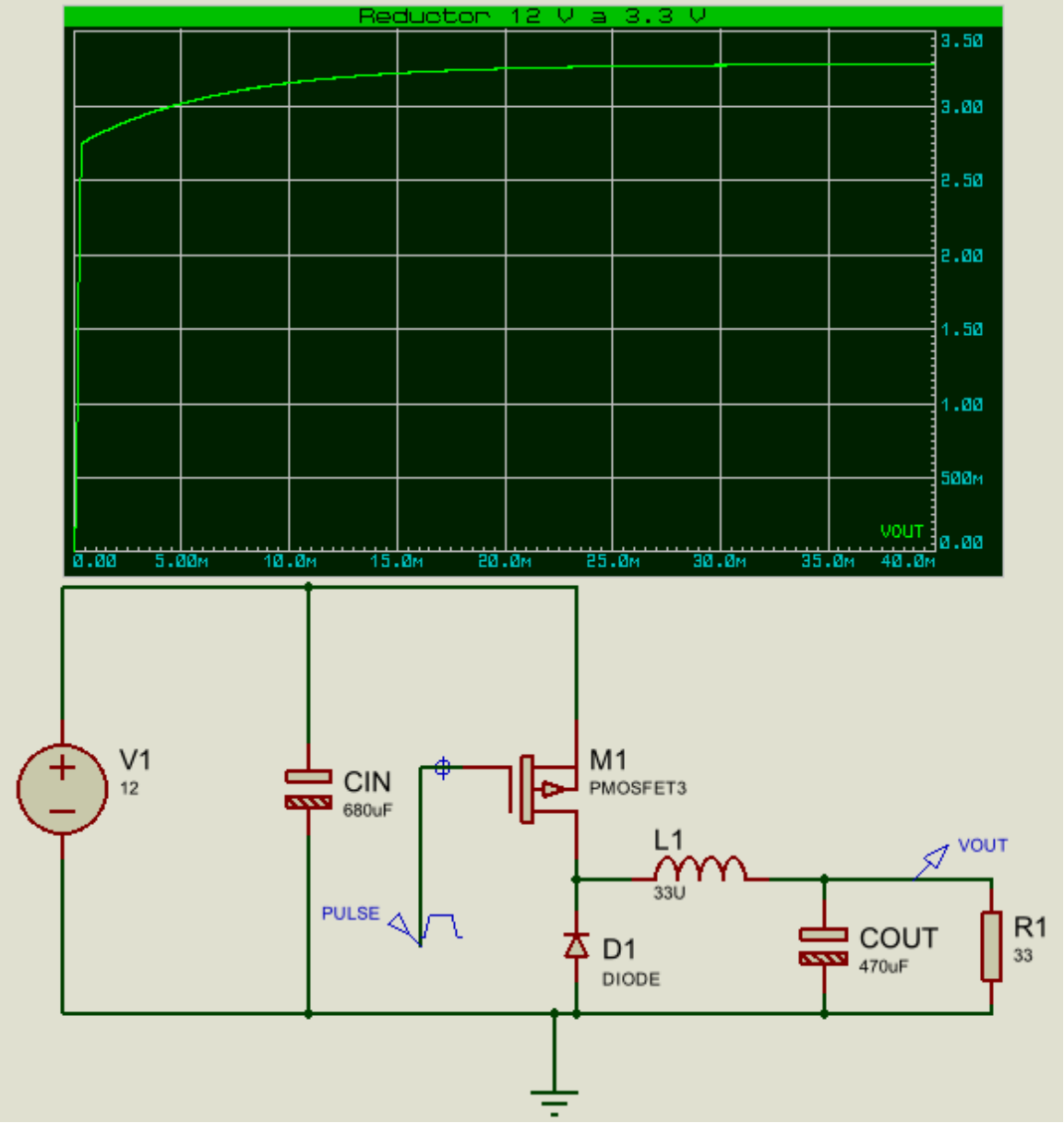


- **D. Dataset details:** CWRU files 97–100 (healthy), 105–108 (inner race), 118–121 (ball), 130–133 (outer race); drive-end accelerometer, 12 kHz, resampled to 13.33 kHz; 6,837 windows of 1,024 samples, 50% overlap.
- **E. Extended model metrics:** confusion matrix on the 3 HP test set (1,052 healthy, 265 inner race, 262 ball, 264 outer race windows; all classified correctly) and the robustness table of Section 5.

F. Proteus VSM firmware simulation: the demo firmware (`firmware/app/ai_sed_proteus.elf`) running on the STM32F411CE of the board. The Virtual Terminal, connected to USART1 (PA9, 115200 baud), shows the diagnosis of 16 test windows from the unseen 3 HP load (clean and with 5 dB noise): "Result: 16/16 correct".



G. Proteus simulation of the buck stage: *Figure 2.* Proteus simulation of the equivalent buck stage: output voltage VOUT (top) for a 12 V input, 150 kHz switching and a 33 Ω load (circuit below), run for 30 ms. VOUT settles at about 3.3 V.


Reductor 12 V a 3.3 V
3.50
3.00
2.50
2.00
1.50
1.00
500M
0.00
VOUT
0.00
5.00M
10.0M
15.0M
20.0M
25.0M
30.0M
35.0M
40.0M
V1
12
CIN
680uF
M1
PMOSFET3
L1
33U
VOUT
PULSE
D1
DIODE
COUT
470uF
R1
33